Intelligent Bionic Robotic Hand

User Manual

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1.Safety Information

Before installing and using the Intelligent Bionic Robotic Hand, please carefully read the information in this chapter. Below are the essential safety usage details for the dexterous hand. This information is intended to protect you, the driver, and related equipment when using the product.

Cautions

Reminder:

• Disconnect the power supply before installation, cleaning, or maintenance.

• Always operate the product within the specified range of parameters.

• Do not use damaged power cables, plugs, or loose sockets, as this may damage the product.

• Avoid touching electrical components to prevent damage from electrostatic discharge.

• Ensure the end effector is correctly and securely mounted and connected via the cable.

• The Intelligent Bionic Robotic hand's power supply current is limited to 2A-3A to prevent excessive current during stalling, which could damage the equipment.

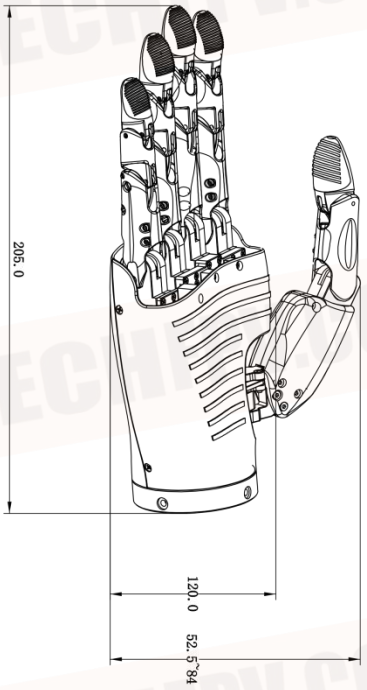
2. Product Introduction

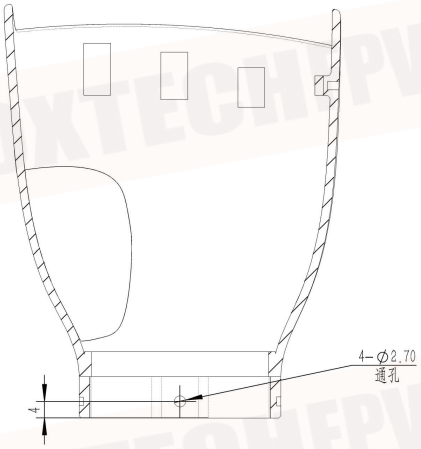
2.1 Product Overview

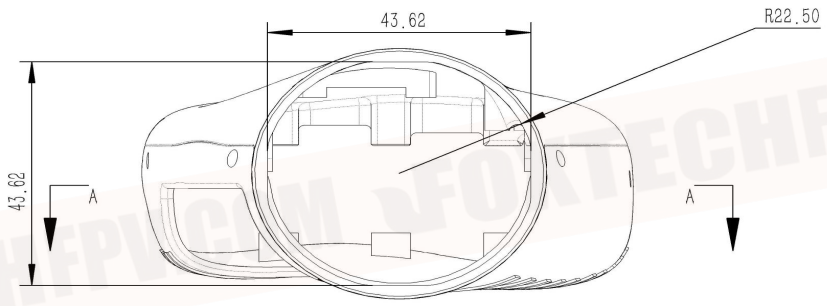
The Intelligent Bionic Robotic Hand is a humanoid robotic hand integrated with 6 actuators, allowing each finger to move independently. It is highly flexible and adaptable, capable of interacting with the surrounding environment, objects, and even humans, while minimizing the risks of injuring the operator, damaging the product, or harming the robot itself. The communication methods used include RS485, Can, and Bluetooth interfaces.

2.2 Mechanical Data

The following diagram shows the overall dimensions of the intelligent bionic hand structure, which is applicable to both the left and right hands. The distance units in the diagram are in millimeters [MM].







2.3 Specifications

Operating Voltage/Current	DC12V/3A
Static Current	200mA
Communication Interface	RS485、Can
Degrees of Freedom	6
Gripping Force	40N
Palm Traction	80N
Maximum Force on Thumb	12N
Force on Each Fingertip	10N
Operating Temperature Range	-20℃～60℃

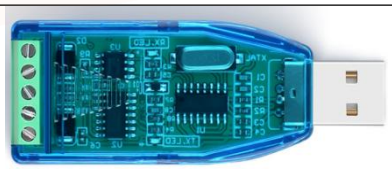
2.4 Finger Codes

The finger codes are the same for both the left and right hands:
Little finger: Code 1, Ring finger: Code 2, Middle finger: Code 3, Index finger: Code 4, Thumb: Code 5, Thumb's metacarpal joint: Code 6

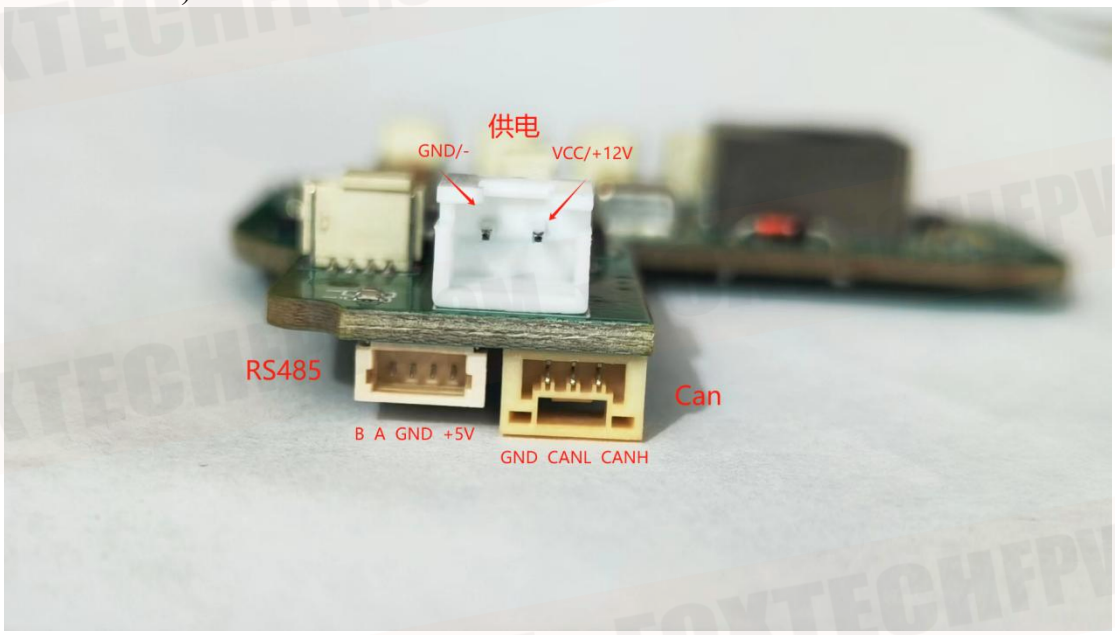




2.5 Electrical Connections

Universal Power Cable	**xh2.54*30cm*2**	
RS485	**USB to RS485*1**	
	MX1.25mm*2	
Can	**X1251H-02-03*1**	

Wiring Definition (Note: The +5V on the RS485 is optional and mainly used to power other devices.)



3. Communication Protocol

3.1Overview of Communication Protocol

RS-485 Communication Protocol

RS-485 (also known as EIA-485) is an industrial standard serial communication protocol used to connect multiple devices. It is a half-duplex communication protocol that allows data transmission between multiple devices, with a maximum of 32 devices connected. One device acts as the master, while the other devices function as slaves.

The advantages of RS-485 technology include its ability to transmit data over long distances, with a maximum distance of 1200 meters. It also supports high transmission speeds, reaching up to 10 Mbps. Additionally, RS-485 is highly resistant to interference, making it capable of withstanding noise and operating in environments with electromagnetic interference (EMI).

RS-485 technology is widely used in various applications, primarily in industrial control systems such as robotic control systems, automation control systems, and smart home systems.

In summary, RS-485 is a highly useful serial communication protocol. Its long transmission distance, high data transfer rate, and strong resistance to interference make it suitable for a range of applications, including industrial control systems, data acquisition systems, intelligent transportation systems, and many other scenarios.

CAN Communication Protocol

The main features of the CAN bus are as follows:

High Reliability: The CAN bus uses differential signaling for data transmission, which provides strong resistance to interference. It also features error detection and recovery mechanisms, significantly enhancing the system's reliability.

Real-Time Capability: The CAN bus utilizes bit arbitration technology, ensuring that high-priority messages are transmitted in a timely manner. This makes it ideal for applications that require real-time responses.

Flexibility: The CAN bus supports various communication speeds, with the highest speed reaching up to 1 Mbps. It also supports a range of data lengths, from 8 bits to 8192 bits, allowing for flexible configuration based on specific needs.

International Standard: The CAN bus is an international standard and is widely used worldwide, facilitating technical exchanges and product interoperability.

3.2 RS485 Communication Protocol

The features of the RS485 protocol are as follows:

- 115200bps Baud rate;
- 1 stop bit;
- 8 data bits;
- No parity check;

Bionic Robotic Hand Parameter Operations Data Type UNS16

03:

FF Power

Byte 0	Byte 1	Byte 2	Byte3	Byte4	Byte 5	Byte 6	Byte7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte14	Byte 15	Byte 16	Byte 17
FF	FF	03	FF	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	FF	FF
Frame Header			Power													Frame Trailer	

A1 Independent Movement Mode

Byte 0	Byte 1	Byte 2	Byte3	Byte4	Byte 5	Byte6	Byte 7	Byte 8	Byte 9	Byte10	Byte11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	03	A1	ID	FF	Acceleration	Target Position		Operation Time		Operation Speed		Torque Limitation		00	FF	FF
Frame Header			Single Movement		/										/	Frame Trailer	

Byte 0	Byte 1	Byte2	Byte3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte17
FF	FF	03	Acceleration 0xA2	ID1		ID2		ID3		ID4		ID5		ID6		FF	FF
Frame Header			Position0xA3													Frame Trailer	
			Time 0xA4														
			Speed 0xA5														
			Torque 0xA6														

B1 Linked Movement Mode

Byte0	Byte 1	Byte 2	Byte 3	Byte4	Byte5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte17
FF	FF	03	B1	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
FrameHeader				Action	补 0											Frame Trailer	

01: All Fingers Open 02: All Fingers Closed (Fist) 03: Extend Index Finger 04: Extend Middle Finger 05: Extend Ring Finger 06: Extend Little Finger 07: Gesture: OK

D1：

0x01 Get Normal Status

Byte 0	Byte 1	Byte 2	Byte3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	01	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Norm-al	ID	Read 1	Volta-ge	Temp	/			Load		Current Position		Status	/	Frame Trailer	

0x02 EEPROM

Byte 0	Byte 1	Byte 2	Byte3	Byte 4	Byte5	Byte6	Byte 7	Byte 8	Byte9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	02	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header			EEPROM	ID	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	/	Frame Trailer	

0x03 Position Debugging

Byte 0	Byte 1	Byte 2	Byte3	Byte 4	Byte 5	Byte6	Byte 7	Byte 8	Byte9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	03	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Position Debugging	ID	Read 1/Write 2	Minimum Angle		Maximum Angle			Correction Position		/	/	/	/	Frame Trailer	

0x04 Unloading Protection

Byte 0	Byte 1	Byte 2	Byte3	Byte4	Byte 5	Byte 6	Byte7	Byte 8	Byte9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	04	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Unloading Protection	ID	Read 1/Write 2	Unloading Condition	/	/	/	/	/	/	/	/	/	/	Frame Trailer	

0x05 Overload Protection

Byte 0	Byte 1	Byte 2	Byte3	Byte4	Byte 5	Byte6	Byte	Byte 8	Byte9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17

							7										
FF	FF	D1	05	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header			Overload Protectio-n	ID	Read 1/Wr ite 2	Prote-ction Torqu-e	/	Prote ction Time	/	Over load Torq-ue	/	/	/	/	/	Frame Trailer	

0x06 Overcurrent Protection

Byte 0	Byte 1	Byte 2	Byte3	Byt e4	Byt e5	Byte 6	Byte 7	Byte 8	By te9	Byt e10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	06	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header			Overcurr-ent Protectio-n	ID	Rea d 1/W r-ite 2	Protection Current		Overc-urrent Time	/	/	/	/	/	/	/	Frame Trailer	

0x07 Working Limitations

Byte 0	Byte 1	Byte 2	Byte3	Byt e4	Byte 5	Byte 6	Byt e7	Byte 8	Byt e9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	07	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header			Working Limitatio ns	ID	Read 1/Wr ite 2	Max Output Torque		Starting Force		Max Temp	/	Max Volt.	/	Mini mum Volt.	/	Frame Trailer	

0x09 Query Current Angle of a Single Finger

byte0	byte1	byte2	byte3	Byte 4	byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	09	01	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header			Single Finger	ID				Angle				Raw Data				Frame Trailer	

0x10 Query Current Angle of a Single Finger

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	10	00	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Debu -g	All	ID1	Ang-le	ID2	Angl -e	ID3	Angle	ID4	Angl -e	ID5	Angl e	ID6	Angl -e	Frame Trailer	

3.3 CAN Communication Protocol

The features of the CAN protocol are as follows:

- **Baud Rate**: 1000 Kbps
- **Frame Format**: Standard Frame
- **Frame Type**: Data Frame

Two-Byte Command

Query Angle:

Byte 0	Byte1
A1	01

Return Values:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
A1	01	ID1Angle	ID2Angle	ID3Angle	ID4Angle	ID5Angle	ID6Angle

Query Version Number:

Byte 0	Byte1
A1	02

Store Parameters to Flash:

Byte 0	Byte1
A1	0E

(This command is used in conjunction with modifying ID and baud rate.)

Action Modes:

Byte 0	Byte1	Function Description
03	01	Open
	02	Fist
	03	Index Finger
	04	Middle Finger
	05	Ring Finger
	06	Little Finger
	07	OK Gesture

Power On/Off：

Byte 0	Byte1	Function Description
03	00	Power Off
03	11	Power On

Modify Device CAN ID:

Byte 0	Byte 1	Function Description
2E	Modified ID	Modify Device CAN ID

Four-Byte Command

Single Finger Position Mode:

Byte 0	Byte1	Byte2	Byte3
04	ID	Angle	00

Single Finger Acceleration Setting:

Byte 0	Byte1	Byte2	Byte3
08	ID	Acceleration	00

Single Finger Speed Setting:

Byte 0	Byte1	Byte2	Byte3
09	ID	Speed	

Single Finger Torque Setting:

Byte 0	Byte1	Byte2	Byte3
10	ID	Torque	

Speed Setting:

Byte 0	Byte1	Byte2	Byte3
05	Speed		00

Torque Setting:

Byte 0	Byte1	Byte2	Byte3
06	Torque		00

Acceleration Setting:

Byte 0	Byte1	Byte2	Byte3
07	Acceleration		00

Baud Rate Setting:

Byte 0	Byte1	Byte2	Byte3
3F	Baud Rate		00

Eight-Byte Command

Position Mode:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
A3	ID1Angle	ID2Angle	ID3Angle	ID4Angle	ID5Angle	ID6Angle	00

Debug

Read Minimum, Maximum, and Correction Values

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
D1	ID	00	00	00	00	00	00

Return Values:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
D1	ID	Servo Minimum Value		Servo Maximum Value		Servo Correction Value	

Write Minimum, Maximum, and Correction Values

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
D2	ID	Servo Minimum Value		Servo Maximum Value		Servo Correction Value	

Read Status

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
D3	ID	00	00	00	00	00	00

Return Values:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
D3	ID	Voltage	Temperature	Current Servo Value		00	00

3.4 Parameter Analysis

Parameter Analysis					
Function	Permission	Minimum Value	Maximum Value	Unit	Value Interpretation
Minimum Position	Read/Write	0	4094	Steps	Sets the minimum position limit for motion travel; value must be less than the maximum angle limit.
Maximum Position	Read/Write	1	4095	Steps	Sets the maximum position limit for motion travel; value must be greater than the minimum angle limit.
Temperature Limit	Read/Write	0	100	°C	Sets the maximum working temperature limit. E.g., setting 70 results in a max temperature of 70°C with a precision of 1°C.
Max Input Voltage	Read/Write	0	254	0.1V	Sets the maximum input voltage limit. E.g., setting 80 limits the voltage to 8.0V with a precision of 0.1V.
Min Input Voltage	Read/Write	0	254	0.1V	Sets the minimum input voltage limit. E.g., setting 40 limits the voltage to 4.0V with a precision of 0.1V.
Max Torque	Read/Write	0	1000	0.10%	Sets the maximum output torque limit. E.g., setting 1000 = 100% stall torque; setting 800 means the stall torque is 40 kg·cm, and the max output is 40 * 0.8 = 32 kg·cm.
Unload Condition	Read/Write	0	254	-	ZL
Position Loop P Coefficient	Read/Write	0	254	-	Proportional Coefficient for Motor Control
Position Loop D Coefficient	Read/Write	0	254	-	Derivative Coefficient for Motor Control
Position Loop I Coefficient	Read/Write	0	254	-	Integral Coefficient for Motor Control
Minimum Starting Force	Read/Write	0	1000	0.1%	Set the minimum output startup torque for the servo. For example, setting 1000 = 100% * stall torque.
Protection Current	Read/Write	0	511	6.5mA	The maximum adjustable current is 500 * 6.5mA = 3250mA.
Position Correction	Read/Write	-2047	2047	Steps	BIT11 is the direction bit, indicating the positive or negative direction. The other bits represent the range from 0 to 2047 steps.
Protection Torque	Read/Write	0	254	1.0%	Output torque after entering overload protection. For

					example, setting 20 means 20% of the maximum torque.
Protection Time	Read/Write	0	254	10ms	The duration of time when the current load exceeds the overload torque and is maintained. For example, setting 200 means 2 seconds, with a maximum setting of 2.5 seconds.
Overload Torque	Read/Write	0	254	1.0%	The torque threshold to trigger overload protection. E.g., setting 80 means 80% of the maximum torque.
Overcurrent Protection Time	Read/Write	0	254	10ms	Maximum possible time is 254 * 10ms = 2540ms.
Torque Switch	Read/Write	0	2	-	Write 0: Turn off torque output; Write 1: Turn on torque output; Write 128: Correct current position to 2048.
Acceleration	Read/Write	0	254	100steps/s^2	E.g., setting 10 results in acceleration/deceleration of 1000 steps per second squared.
Target Position	Read/Write	-32766	32766	Steps	Each step represents the smallest resolvable angle, using absolute position control mode, where the maximum corresponds to the maximum effective angle.
Operation Time	Read/Write	0	1000	0.10%	
Operating Speed	Read/Write	0	254	steps/s	The number of steps moved per unit time (per second). For example, 50 steps/second = 0.732 RPM (revolutions per minute).
Torque Limit	Read/Write	0	1000	1.0%	The initial value upon power-up is set by the maximum torque (0x10), and users can programmatically modify this value to control the maximum torque output.
Current Position	Read-Only	-1	-1	steps	Feedback the current position in terms of steps, where each step represents the smallest resolvable angle. In absolute position control mode, the maximum value corresponds to the maximum effective angle.
Current Speed	Read-Only	-1	-1	steps/s	Feedback the current speed of the motor, expressed as the number of steps moved per unit time (per second).
Current Load	Read-Only	-1	-1	0.1%	The current duty cycle of the

					voltage output driving the motor.
Current Voltage	Read-Only	-1	-1	0.1V	The current operating voltage of the servo.
Current Temperature	Read-Only	-1	-1	°C	The current internal operating temperature of the servo.
Servo Status	Read-Only	-1	-1	-	The corresponding bit is 0 for no error, and 1 for an error. Bit0: Voltage Bit1: Sensor Bit2: Temperature Bit3: Current Bit4: Angle Bit5: Overload
Movement Flag	Read-Only	-1	-1	-	The servo is marked as 1 when in motion and 0 when stopped.
Current Current	Read-Only	-1	-1	6.5mA	The maximum measurable current is 500 * 6.5mA = 3250mA.

4. Version Correction

Version Update: Bug fix descriptions, optimization function descriptions.

Version Bug: A bug was found in the version, marked in red to highlight the bug's version source, with the cause.

4.1 RS485 Version Correction

2023.07.31

Added two commands for querying the angle:

1. Query the current angle of a single finger: 0x09; Data type: UNS16

Send:

FF FF D1 09 01 00 00 00 00 00 00 00 00 00 00 FF FF

Parsing:

byte0	byte 1	byte2	byte3	byte 5	Byte6	Byte7	Byte8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	09	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Debug	Single Finger	Padding											Frame Trailer	

Return:

FF FF D1 09 01 00 00 00 2C 00 00 00 3B 09 00 00 FF FF

Parsing:

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	09	01	00	00	00	2C	00	00	00	3B	09	00	00	FF	FF
Frame Header		Debug	Single Finger	ID				Angle				Raw Data				Frame Trailer	

Algorithm: X = (Maximum value - Raw data) * 90 / (Maximum value - Minimum value)
Note: X is the angle value (range 0°～90°); Raw data refers to the values obtained from Byte[12] and Byte[13]; Maximum and minimum values are the limit values in the Debug settings. The obtained data is in hexadecimal and should be converted to decimal for calculation.

2. Query All Finger Current Angles 0x10; Data Type UNS16
FF FF D1 10 00 00 00 00 00 00 00 00 00 00 00 FF FF
Return:

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15	Byte 16	Byte 17
FF	FF	D1	10	00	00	00	00	00	00	00	00	00	00	00	00	FF	FF
Frame Header		Debug	All	ID1	Angle	ID2	Angle	ID3	Angl -e	ID4	Angl -e	ID5	Angl -e	ID6	Angl -e	Frame Trailer	